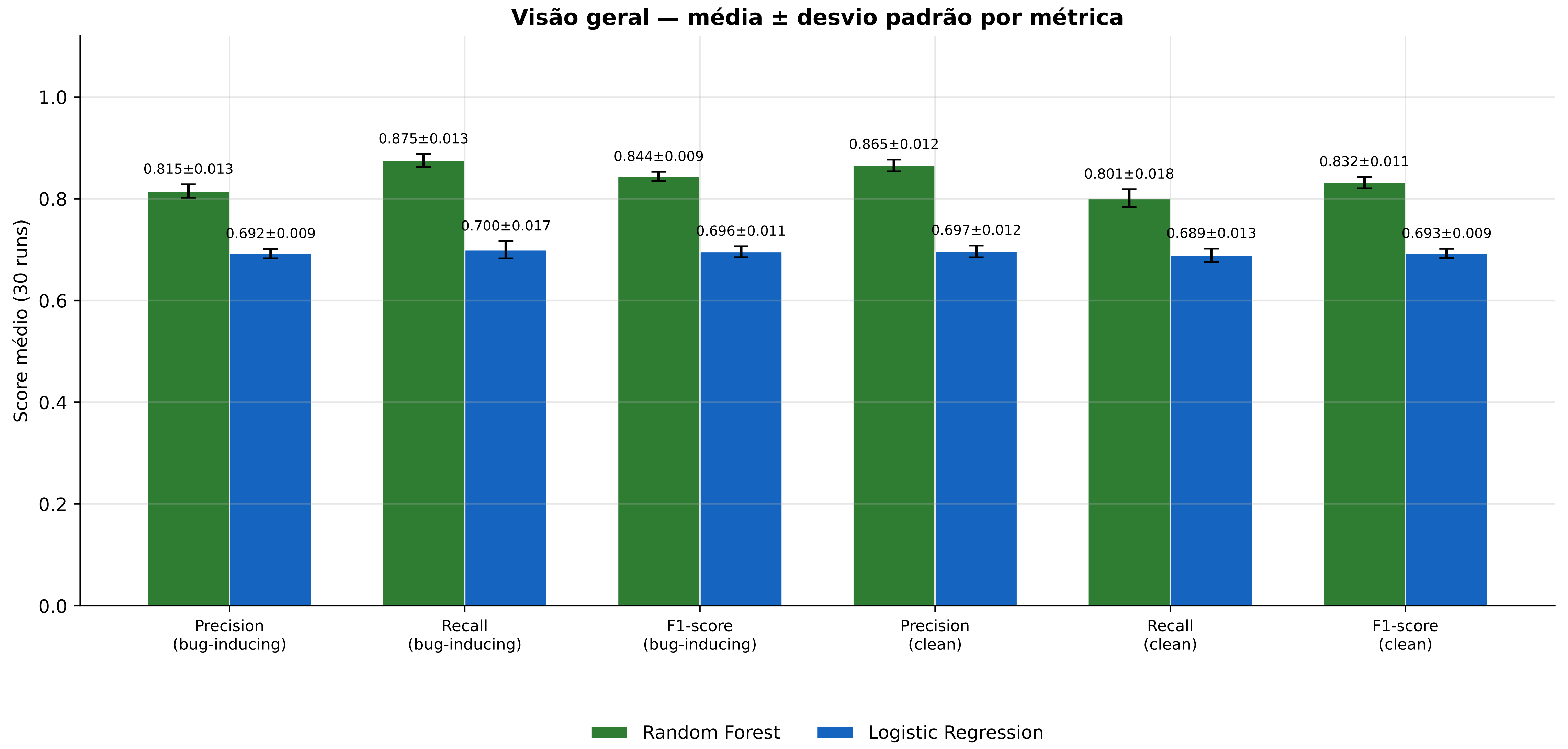
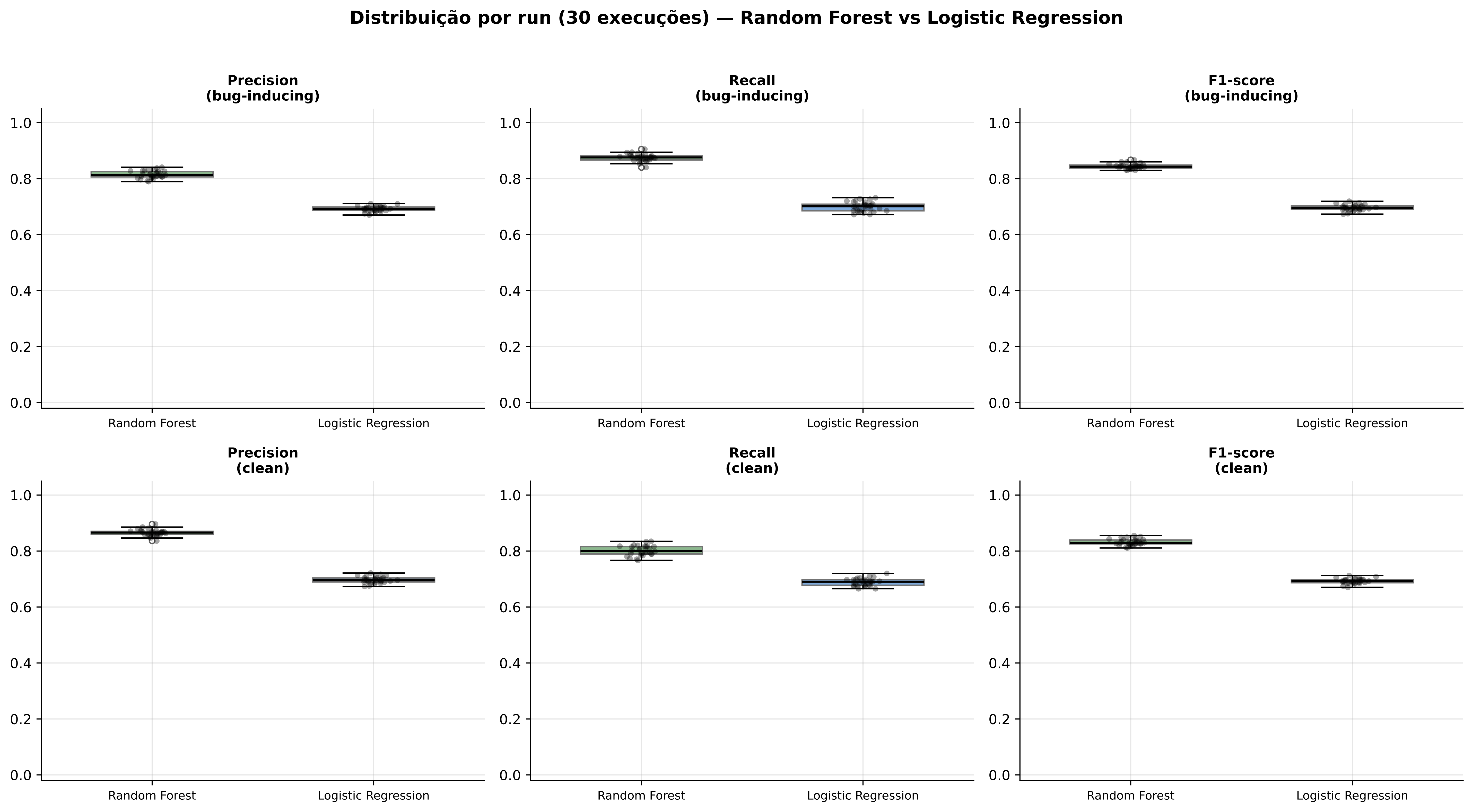
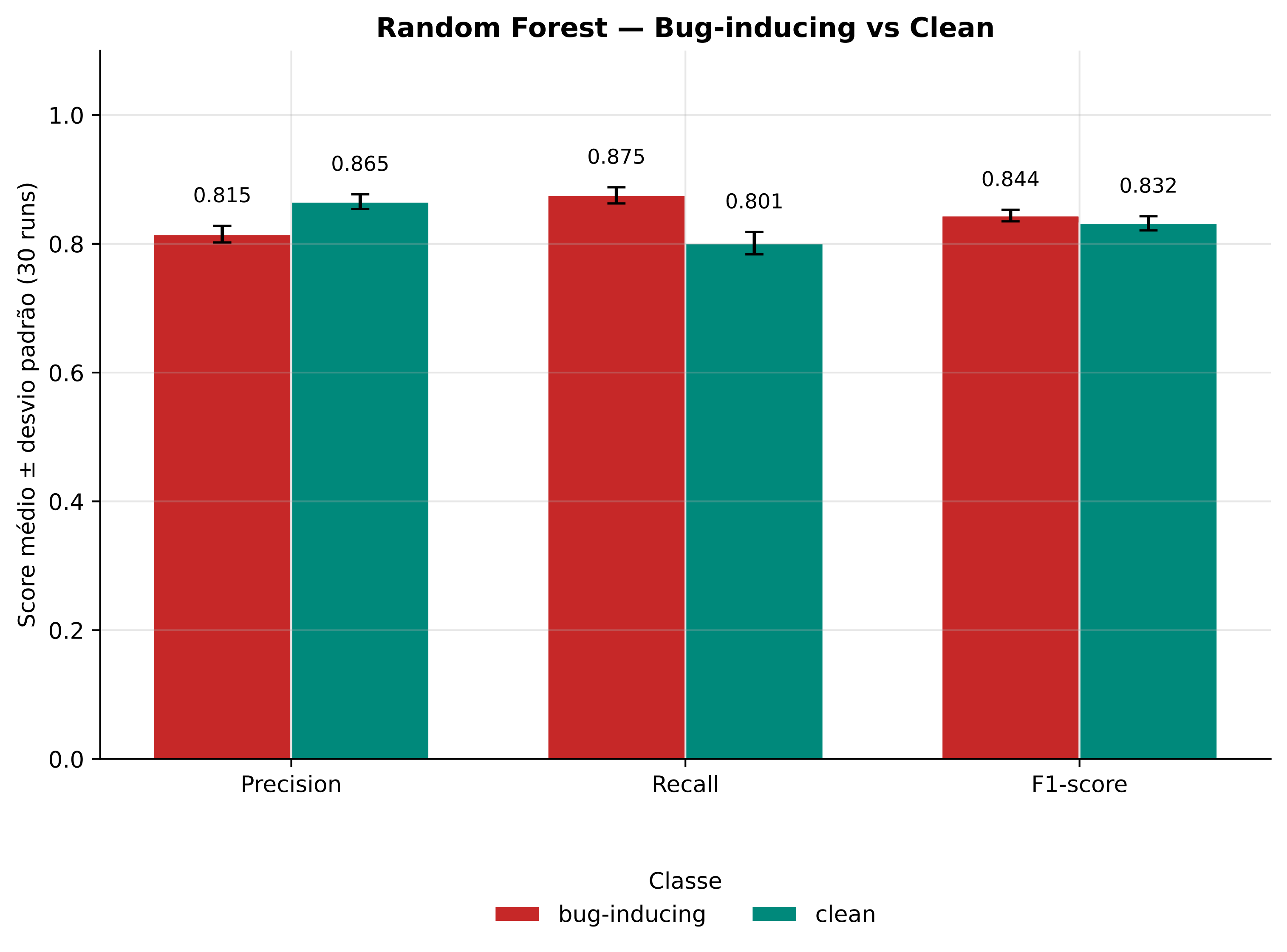
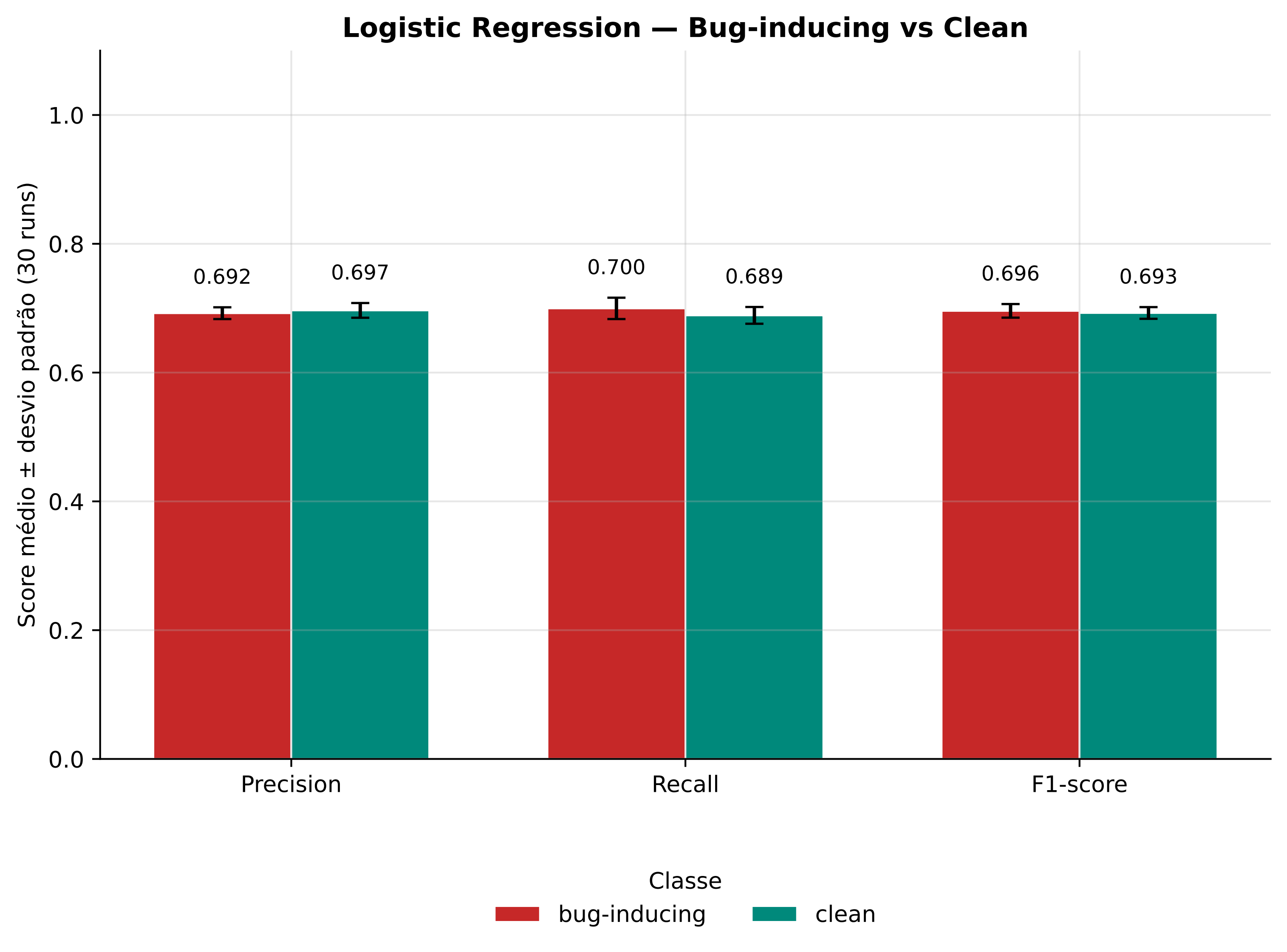
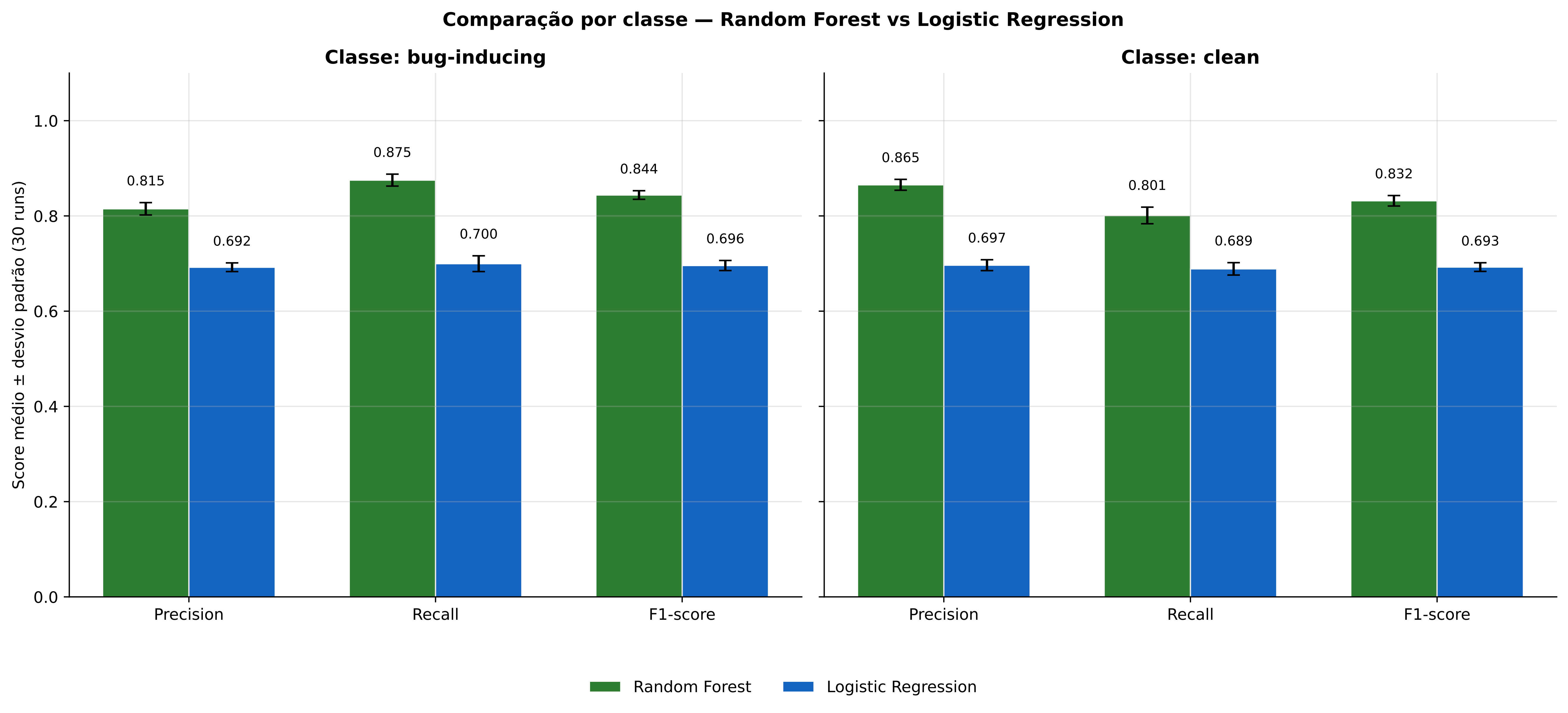

Visão geral — média ± desvio padrão por métrica
Score médio (30 runs)
1.0
0.8
0.6
0.4
0.2
0.0
0.815±0.013
0.692±0.009
0.875±0.013
0.700±0.017
0.844±0.009
0.696±0.011
0.865±0.012
0.697±0.012
0.801±0.018
0.689±0.013
0.832±0.011
0.693±0.009
Precision
(bug-inducing)
Recall
(bug-inducing)
F1-score
(bug-inducing)
Precision
(clean)
Recall
(clean)
F1-score
(clean)
Random Forest
Logistic Regression


Distribuição por run (30 execuções) — Random Forest vs Logistic Regression
Precision (bug-inducing)
Recall (bug-inducing)
F1-score (bug-inducing)
Precision (clean)
Recall (clean)
F1-score (clean)
0.0
0.2
0.4
0.6
0.8
1.0
Random Forest
Logistic Regression

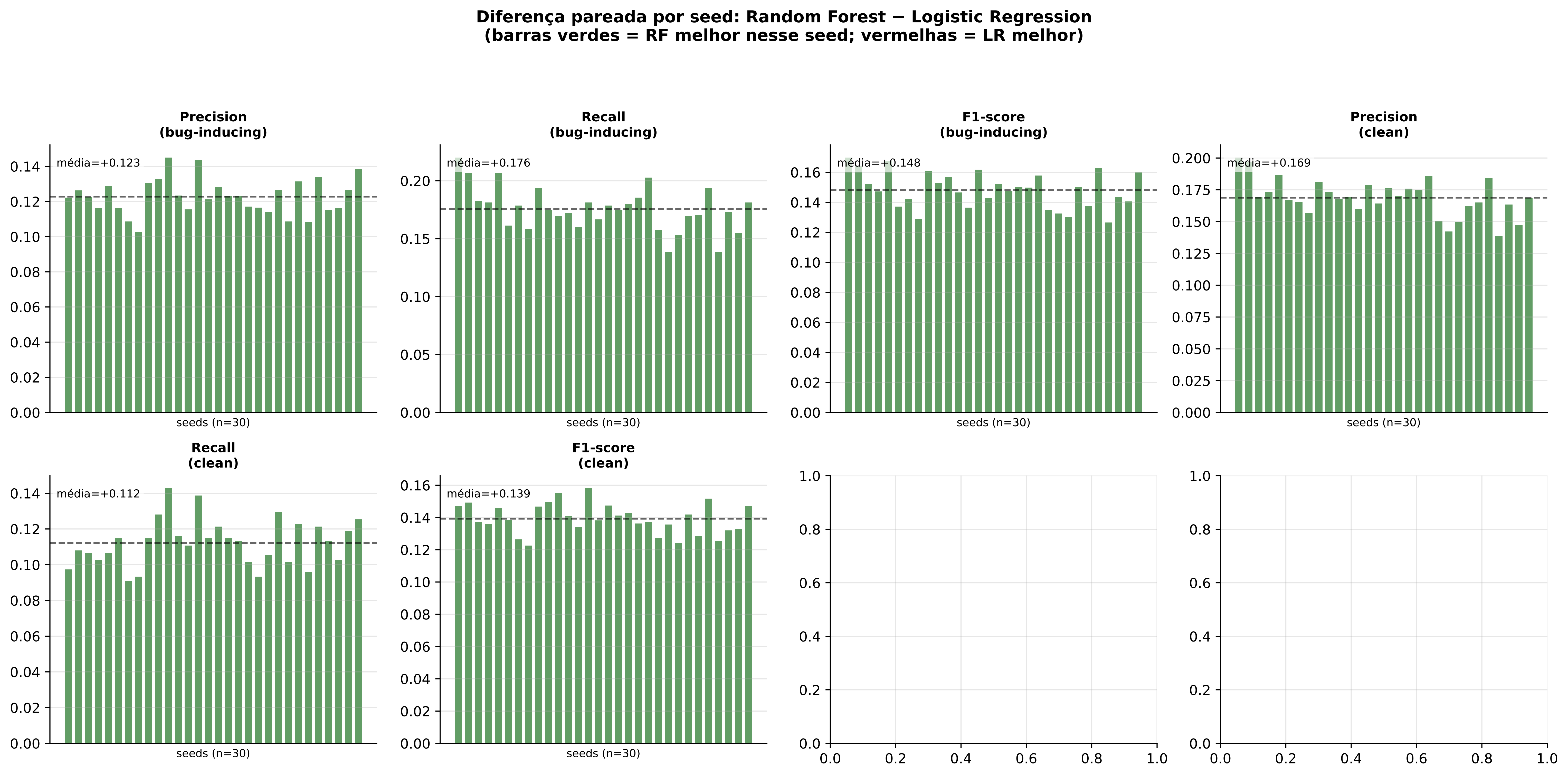


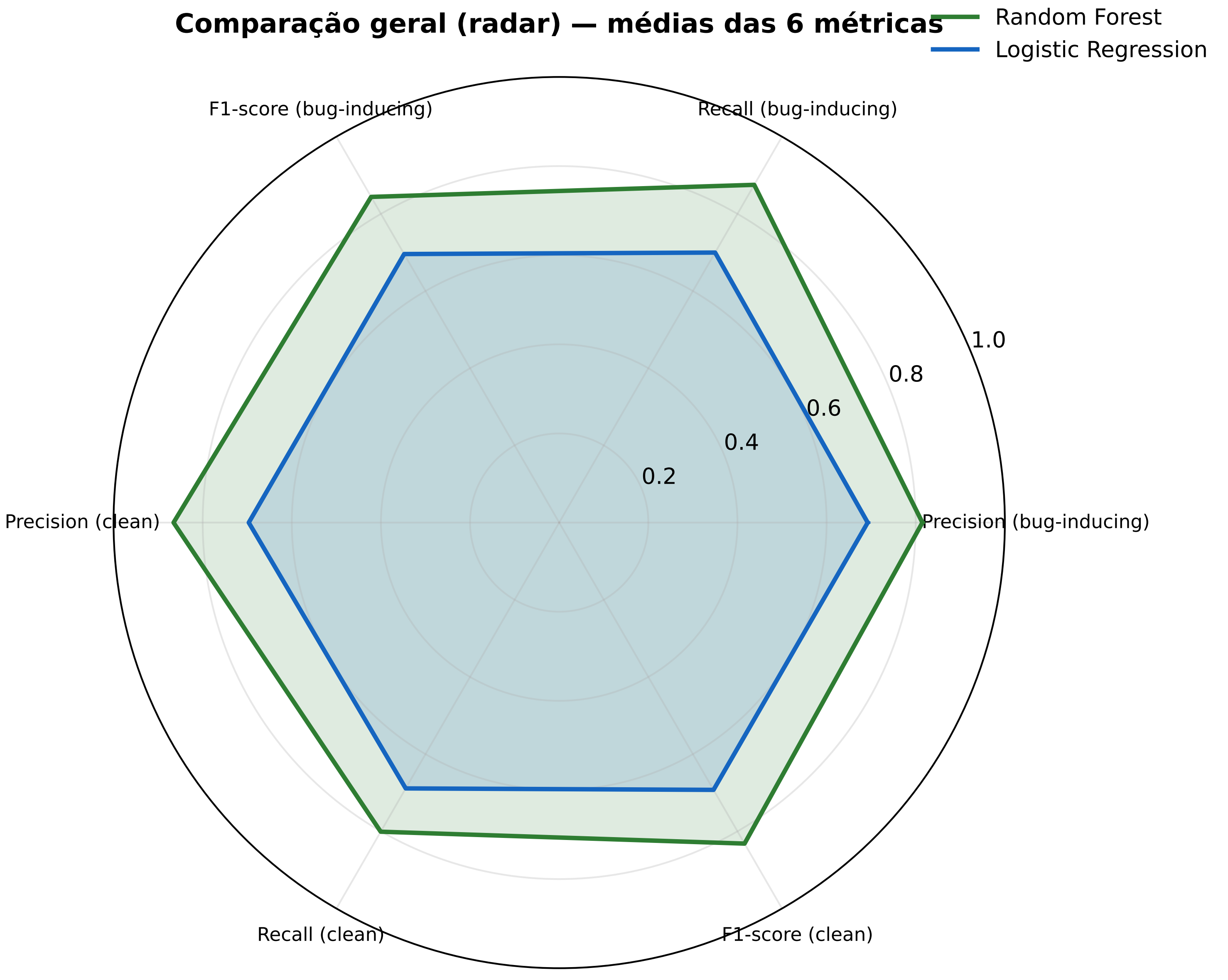
Random Forest — Bug-inducing vs Clean
Score médio ± desvio padrão (30 runs)
1.0
0.8
0.6
0.4
0.2
0.0
0.815
0.865
0.875
0.801
0.844
0.832
Precision
Recall
F1-score
Classe
bug-inducing
clean


Logistic Regression — Bug-inducing vs Clean
Score médio ± desvio padrão (30 runs)
1.0
0.8
0.6
0.4
0.2
0.0
0.692
0.697
0.700
0.689
0.696
0.693
Precision
Recall
F1-score
Classe
bug-inducing
clean

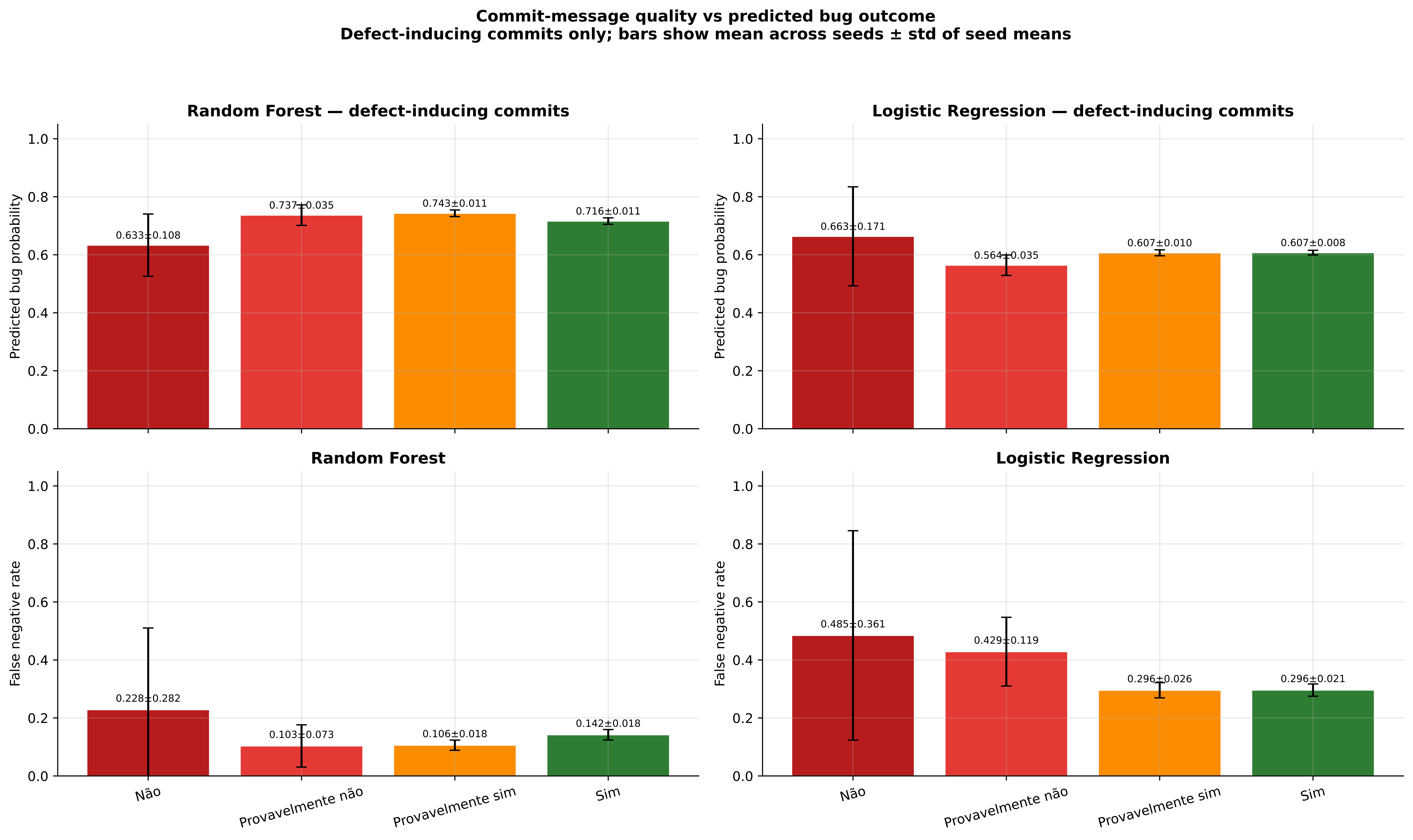


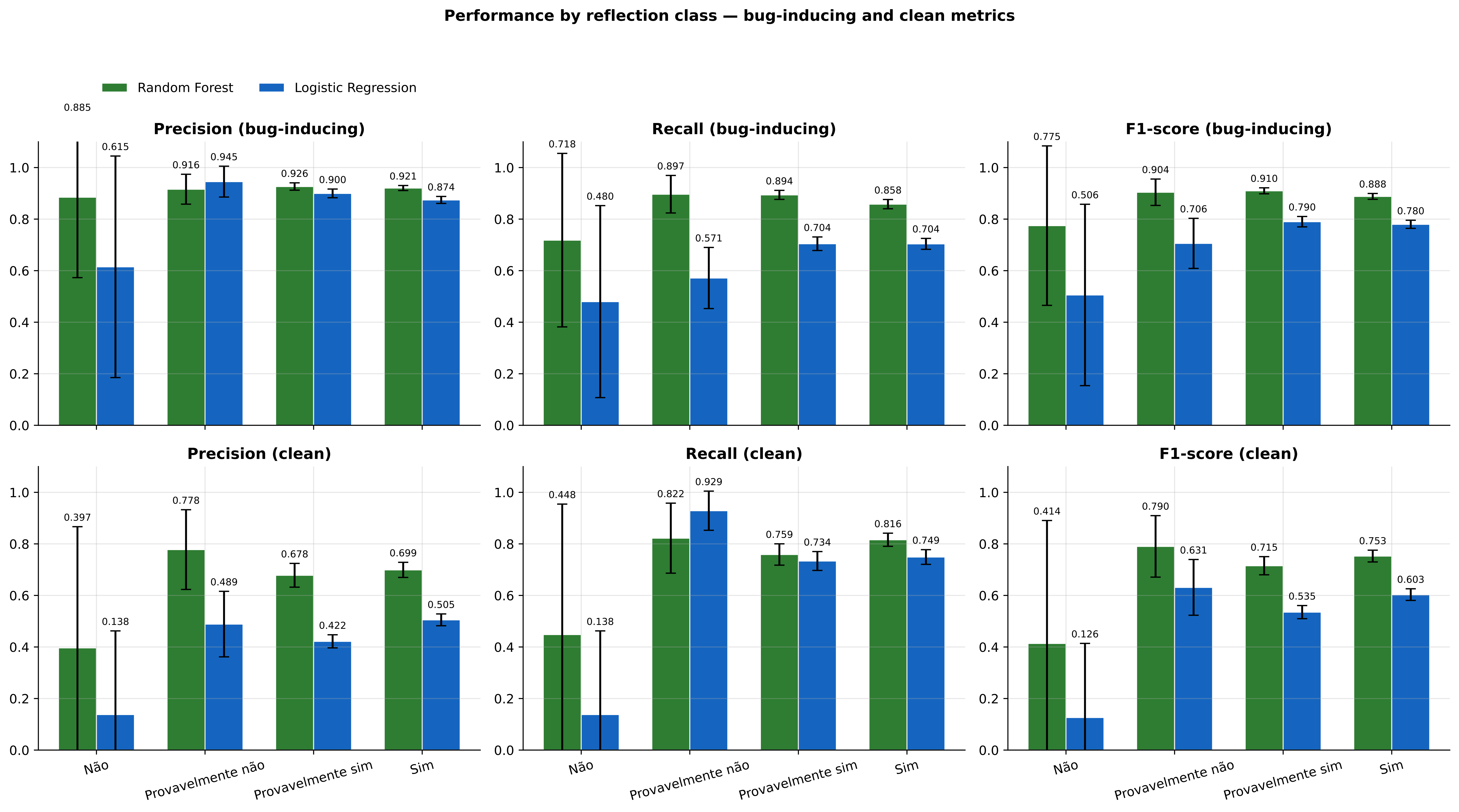
Comparação por classe — Random Forest vs Logistic Regression
Classe: bug-inducing
Classe: clean
Score médio ± desvio padrão (30 runs)
0.815
0.692
0.875
0.700
0.844
0.696
0.865
0.697
0.801
0.689
0.832
0.693
Precision
Recall
F1-score
Random Forest
Logistic Regression


Diferença pareada por seed: Random Forest − Logistic Regression
(barras verdes = RF melhor nesse seed; vermelhas = LR melhor)
Precision (bug-inducing)
média=+0.123
Recall (bug-inducing)
média=+0.176
F1-score (bug-inducing)
média=+0.148
Precision (clean)
média=+0.169
Recall (clean)
média=+0.112
F1-score (clean)
média=+0.139
seeds (n=30)



Comparação geral (radar) — médias das 6 métricas
Random Forest
Logistic Regression
F1-score (bug-inducing)
Recall (bug-inducing)
Precision (clean)
Precision (bug-inducing)
Recall (clean)
F1-score (clean)
0.2
0.4
0.6
0.8
1.0


Commit-message quality vs predicted bug outcome
Defect-inducing commits only; bars show mean across seeds ± std of seed means
Random Forest — defect-inducing commits
Logistic Regression — defect-inducing commits
Predicted bug probability
0.633±0.108
0.737±0.035
0.743±0.011
0.716±0.011
0.663±0.171
0.564±0.035
0.607±0.010
0.607±0.008
Random Forest
Logistic Regression
False negative rate
0.228±0.282
0.103±0.073
0.106±0.018
0.142±0.018
0.485±0.361
0.429±0.119
0.296±0.026
0.296±0.021
Não
Provavelmente não
Provavelmente sim
Sim



Performance by reflection class — bug-inducing and clean metrics
Random Forest
Logistic Regression
Precision (bug-inducing)
0.885
0.615
0.916
0.945
0.926
0.900
0.921
0.874
Recall (bug-inducing)
0.718
0.480
0.897
0.571
0.894
0.704
0.858
0.704
F1-score (bug-inducing)
0.775
0.506
0.904
0.706
0.910
0.790
0.888
0.780
Precision (clean)
0.397
0.138
0.778
0.489
0.678
0.422
0.699
0.505
Recall (clean)
0.448
0.138
0.822
0.929
0.759
0.734
0.816
0.749
F1-score (clean)
0.414
0.126
0.790
0.631
0.715
0.535
0.753
0.603
Não
Provavelmente não
Provavelmente sim
Sim